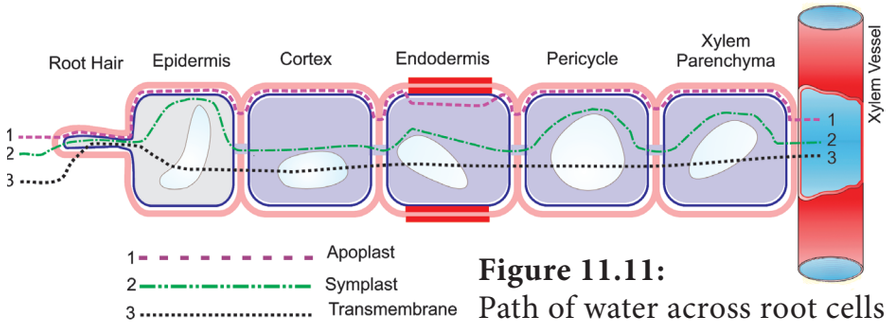


Figure 11.11:
Path of water across root cells

to the plasma membrane of the living cell. The apoplast includes cell walls, extra cellular spaces and the interior of dead cells such as vessel elements and tracheids. In the apoplast pathway, water moves exclusively through the cell wall or the non-living part of the plant without crossing any membrane. The apoplast is a continuous system.

2. Symplast

The **symplast** (Greek: ***sym*** = within; ***plast*** = cell) consists of the entire mass of cytosol of all the living cells in a plant, as well as the **plasmodesmata**, the cytoplasmic channel that interconnects them.

In the symplastic route, water has to cross plasma membrane to enter the cytoplasm of outer root cell; then it will move within adjoining cytoplasm through plasmodesmata around the vacuoles without the necessity to cross more membrane, till it reaches xylem.

3. Transmembrane route

In transmembrane pathway water sequentially enters a cell on one side and exits from the cell on the other side. In this pathway, water crosses at least two membranes for each cell. Transport across the **tonoplast** is also involved.

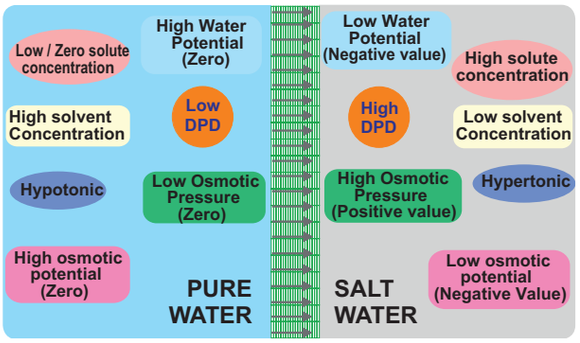
11.4.3 Mechanism of Water Absorption

Kramer (1949) recognized two distinct mechanisms which independently operate in the absorption of water in plants. They are, i) active absorption ii) passive absorption.

1. Active Absorption

The mechanism of water absorption due to forces generated in the root itself is called **active absorption**. Active absorption may be osmotic or non-osmotic.

Concept Map - Movement of water in an osmotic system based on various parameters



i. Osmotic active absorption

The theory of osmotic active absorption was postulated by **Atkins** (1916) and **Preistley** (1923). According to this theory, the first step in the absorption is soil water imbibed by cell wall of the root hair followed by osmosis. The soil water is hypotonic and cell sap is hypertonic. Therefore, soil water diffuses into root hair along the concentration gradient (endosmosis). When the root hair becomes fully turgid, it becomes hypotonic and water moves osmotically to the outer most cortical cell. In the same way, water enters into inner cortex, endodermis, pericycle and finally reaches protoxylem. As the sap reaches the protoxylem a pressure is developed known as **root pressure**. This theory involves the symplastic movement of water.

Objections to osmotic theory: 1.The cell sap concentration in xylem is not always high. 2. Root pressure is not universal in all plants especially in trees.

ii. Non-Osmotic active absorption

Bennet-Clark (1936), **Thimann** (1951) and **Kramer** (1959) observed absorption of water even if the concentration of cell sap in the root hair is lower than that of the soil water. Such a

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